

Mathematics Research Paper

Title: Approximating Pi (π) using Gregory-Leibniz Series and Monte Carlo methodology.

RQ: To what extent do the Gregory-Leibniz series and a quarter-circle Monte Carlo estimator approximate π ?

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Approximating Pi (II) using Gregory-Leibniz Series, Monte Carlo method, and error analysis.

Introduction

1.1 The Pi constant.

Pi (π), known as the representation of the ratio of the circumference and the diameter of a circle. It is a constant that has been used for several centuries now in the symmetrical and geometrical aspect of mathematics. The constant π is vital in numerous real life applications such as engineering, physics, medicine, and the most commonly used is the navigation systems. My interest in the approximation of π was gradually growing as I was familiarized with topics such as geometry in elementary school and trigonometry in high school. As I was introduced to this precise constant calculated as 3.1415926535.. is a continuously recurring number which is due to its irrationality. The idea that π can never be expressed exactly as a particular fraction is what boosted my curiosity behind the logical implementation of mathematicians historically to approximate this value. While studying the Maclaurin series in the AA HL syllabus, I realized that substituting x as 1 in the $\arctan(x)$ expansion directly produces an infinite alternating sum resulting as $\frac{\pi}{4}$. This is what motivated me to investigate how effectively this series actually approximates π , and with a probabilistic Monte Carlo simulation, I can compare the accuracy of the two approaches.

1.2 Mathematical approaches

1.2.1 Gregory-Leibniz Series.

In the further investigation I will derive the Maclaurin series for $\arctan(x)$ and by evaluating at $x = 1$, I can successfully obtain the Gregory-Leibniz series as $(\frac{\pi}{4})$. The Gregory-Leibniz series provides us with a comprehensive idea about how the summation provides us with an amazing alternating series (+,-,+,-) and essentially underpins the approximation of π .

1.2.2 The Monte Carlo approach

As a comparative validation I will also employ the use of the Monte Carlo “dartboard” method, where I will be sampling N random input points on the quarter square circle. It is an approximation approach that significantly uses controlled randomness that estimates a value when an exact formula is inaccurate or impractical. In the following investigation I aim to use the dartboard simulation method to estimate π .

In the following image (figure 1) it shows a quarter circle embedded within a square, this means it is a unit square with a circle inside it. Many random points are generated with the x and y coordinates ranging between 0 and 1 as each location in the square is equally likely. This specific proportion of points that fall inside the arc shows how much of the square the quarter circle occupies. By scaling that proportion successfully, it will provide an estimated value of π . Instead of focusing on pure algebra, using this computationally-possible method, I can easily infer the approximation of π , by well-distributed random samples.

2.0 Research question

The research question states that “**To what extent do the Gregory-Leibniz series and a quarter-circle Monte Carlo estimator approximate π ?**” .

2.1 Rationale

The reason why I have selected the Gregory series is because of its elegant derivation and famous for its result in the Leibniz formula. This Internal Assessment significantly aims to show the core Maths AA HL skills such as convergence proofs, quantitative analysis of the formula’s primary limitation. With additional out-of-syllabus content, I have also considered the use of Monte Carlo simulations as probabilistic models, which allow a computational understanding of the π value. Hence by the use of the 2 methodologies, I will be able to determine the difference and accuracy between the 2 different approaches, theoretically and probabilistically.

2.1.1 Error checking analysis and evaluation:

Once the approximation of π is successfully achieved, I will be considering the error checking analysis between the 2 approximation strategies, and evaluate which one is more approachable, accurate and more reasonable to use. This provides a detailed understanding of how computational and probabilistic approximations could contrast between theoretically implemented mathematics.

3.0 Deriving the Gregory-Leibniz Series from $\arctan(x)$

To achieve the Gregory-Leibniz series, the Maclaurin series of the function $f(t)$ is first required to be derived. Since $\arctan(x)$ is the quintessential integral of $f(t)$, its Maclaurin expansion can then be found. Substituting x as 1, hence yields the Gregory-Leibniz series for π

The exclusive derivation formula for the Maclaurin series is as follows:

$$f(t) = \frac{1}{1+t^2}$$

Setting the **common ratio** as $r = -t^2$ with the **first term** $a = 1$

$$S = \sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$$

$$\text{for } |r| < 1$$

$$\frac{1}{1+t^2} = \frac{1}{1-(-t^2)}$$

$$\frac{1}{1-(-t^2)} = \sum_{n=0}^{\infty} (-t^2)^n$$

$$\sum_{n=0}^{\infty} (-t^2)^n = \sum_{n=0}^{\infty} (-1)^n t^{2n}$$

The Maclaurin series expansion for f(t) in the expanded form would essentially be:

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - t^6 + t^8 - \dots$$

This series successfully **converges** for the interval $| -t^2 | < 1$

3.1 Derivation of the Maclaurin series for the function arctan(x).

The arctan(x) function is defined as a specific definite integral of the function

$$f(x) = \int_0^x \frac{1}{1+t^2} dt$$

To derive the final Maclaurin series for the arctan(x) function, I must consider the fundamental theorem and the identity in order to integrate the power series term by term by using summation, while also seeing that the series is converging with a determined convergence interval:

$$\arctan(x) = \int_0^x \left(\sum_{n=0}^{\infty} (-1)^n t^{2n} \right) dt$$

$$\int_0^x \left(\sum_{n=0}^{\infty} (-1)^n t^{2n} \right) dt = \sum_{n=0}^{\infty} (-1)^n \int_0^x t^{2n} dt$$

To further evaluate the general term, I will be using the **power rule**:

$$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

The final expanded form of the Maclaurin series of $\arctan(x)$:

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

Based on the obtained expanded form of the Maclaurin series for $\arctan(x)$, I can finally transition to the Leibniz formula for π .

$$\arctan(1) = \frac{\pi}{4} :$$

$$\arctan(1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} (1)^{2n+1}$$

This substitution of the $\arctan(1)$ identity provides with the Leibniz Formula:

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \frac{1}{13} - \frac{1}{15} \dots$$

Making π as the subject of the formula by multiplying 4 to the RHS:

$$\left(\frac{\pi}{4}\right) * 4 = 4 * \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \frac{1}{13} - \frac{1}{15} \dots\right)$$

$$\cancel{\left(\frac{\pi}{4}\right)} * \cancel{4} = 4 * \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \frac{1}{13} - \frac{1}{15} \dots\right)$$

$$\pi = \left(4 - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \frac{4}{9} - \frac{4}{11} + \frac{4}{13} - \frac{4}{15} \dots\right)$$

$$\pi \approx 3.14159265359$$

3.2 Alternating Series Test (Proofs)

For the Leibniz series to converge there are 2 conditions that must be met. **Firstly** the alternating Leibniz series must monotonically decrease in its absolute value to show the convergence. **Second**, is where the series' value must approach zero in the limit. So to validate whether the series is converging, I will be taking a look at the authentic Leibniz alternating series invention.

3.2.1 Condition 1: Monotonically decreasing series

The original alternating series was discovered by Gottfried Leibniz, where he displayed it using the summation format:

$$\sum_{n=0}^{\infty} (-1)^n a_n = a_0 - a_1 + a_2 - a_3 + \dots$$

Now to prove that this series converges I must demonstrate the proof by using the **Monotonic decreasing proof** method.

Monotonically decreasing series is a series where each term is less than the previous term. A short representation of the series:

$$a_{n+1} < a_n$$

Proofs for the foundational formula are displayed below.

Proof 1:

$$a_n - a_{n+1} = \frac{1}{2n+1} - \frac{1}{2n+3}$$

$$\frac{1}{2n+1} - \frac{1}{2n+3} = \frac{(2n+3)-(2n+1)}{(2n+1)(2n+3)} = \frac{2}{(2n+1)(2n+3)} > 0$$

$$a_n - a_{n+1} > 0$$

$$a_n > a_{n+1}$$

$$a_{n+1} < a_n$$

Proof 2:

$$\frac{a_{n+1}}{a_n} = \frac{2n+1}{2n+3} < 1 \Rightarrow a_{n+1} < a_n$$

Based on the proofs, I can justify that the following $a_{n+1} < a_n$ formula is applicable to the largest value of n.

3.2.2 Condition 2: Limit condition

$$\lim_{n \rightarrow \infty} a_n = 0$$

As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$ proportionally due to the limit condition, and therefore $\frac{1}{2n+1} \rightarrow 0$

4.0 Detailed visualization for converging series.

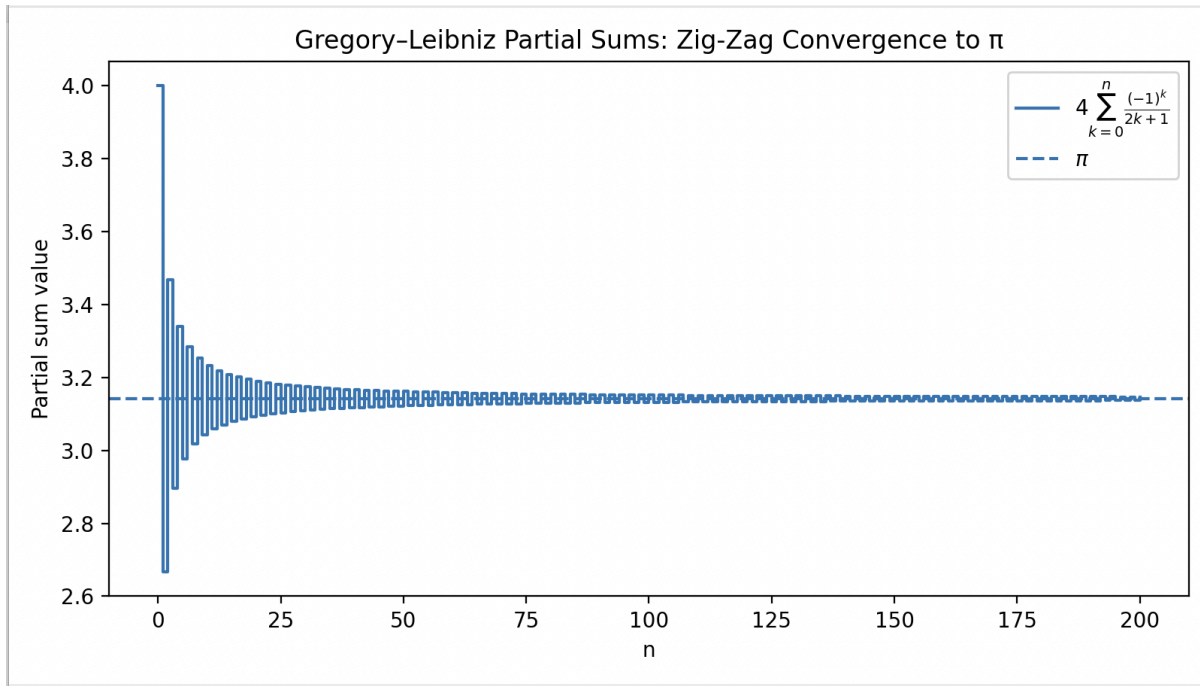
4.1 Zig zag convergence to π

The derived formula for the Gregory - Leibniz series:

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

To be able to visually determine a successful yet accurate convergence of the Gregory-Leibniz series, I utilized my knowledge in the Python programming language. This allows me to generate a graph with varying values of n, proving that the Gregory-Leibniz series successfully converges

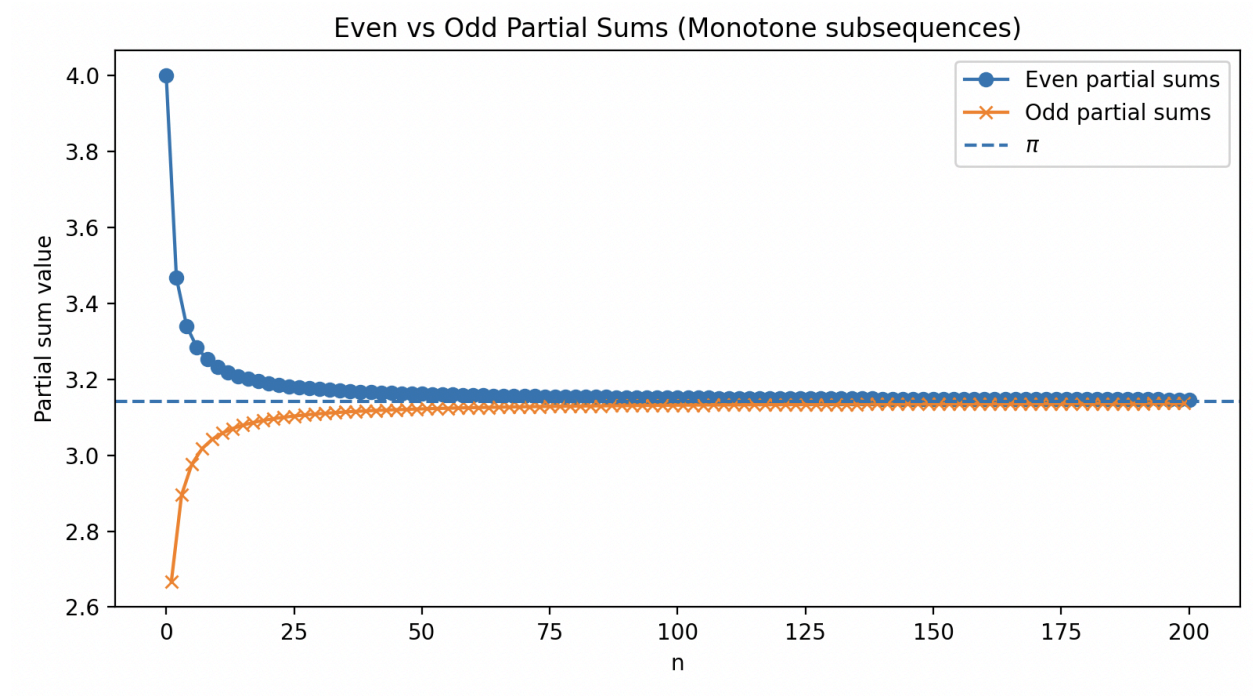
due to its alternating series.



Graph 1 programmatically and visually proves that the Gregory - Leibniz's alternating series is a gradually converging series toward π . (See Appendix 1)

4.2 Odd and Even Sequence

Both the positive and negative sums of the alternating series mathematically tend to increase and decrease. Despite this alternating sign, they still converge to the same constant sum as it is a requirement for both positive and negative sums. The sequence begins with the odd partial sums ($S_1, S_3, S_5, \dots$). These odd partial sums provide the decreasing segment of the sequence, which means that each sum is certainly less than ($<$) the previous term. The even partial sums of the sequence ($S_2, S_4, S_6, \dots$), begins with the sum of the first 2 sums, and then infinitely includes every other subsequent partial sum of the Gregory-Leibniz alternating sequence. Additionally due to the increasing sequence, the second term is always greater than the previous one, $S_4 > S_3$. The even sequence sums are all bounded above the first odd partial sum whereas in the odd sequence, it is bounded below the first even partial sum. As the odd sequence decreases whilst being below the first even partial sum (S_2), and as the even sequence increases whilst being above the first odd partial sum (S_1), both sides of the sequence (+ and -) converge to a limit. The following figure, which is generated by the **self-written python code**, describes the odd and even sums of the alternating series, and shows how effectively each sum converges towards π with varying n-values from both odd and even sums.



Graph 2: Graph that demonstrates the positive and negative converging sums. (See Appendix 2)

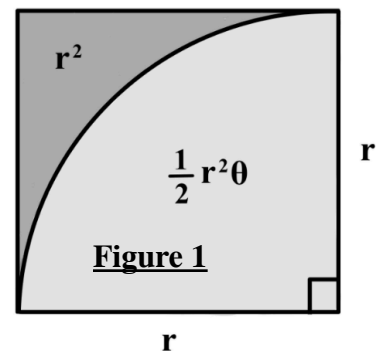
The **Graph 2** shows the behaviour of the odd and even partial sums of the Gregory–Leibniz series. As the number of terms increases, both subsequences move closer to the value of π , demonstrating the characteristic alternating “zig-zag” convergence of the series.

5.0 Evaluating the Monte Carlo method.

A small derivation to prove the quarter of the circle represents $\frac{\pi}{4}$:

Area to probability connection:

Area (A) represents the quarter circle of radius from figure 1, and Area (S) represents the unit square.



$$p = \frac{\iint_A 1dA}{\iint_S 1dA} = \frac{\text{Area}(A)}{\text{Area}(S)}$$

Approach 1:

$$\begin{aligned} \text{Area (A)} &= \int_0^{\frac{\pi}{2}} \int_0^1 r dr d\theta = \left[\frac{1}{2} r^2 \right]_0^1 \left[\theta \right]_{\frac{\pi}{2}}^{\pi} = \frac{1}{2} \left(\pi - \frac{\pi}{2} \right) \\ \mathbf{P} &= \frac{1}{2} \left(\frac{\pi}{2} \right) \\ \mathbf{P} &= \frac{\pi}{4} \end{aligned}$$

Approach 2:**Area (A) :**

$$\begin{aligned}\frac{1}{2}r^2\Theta &= \frac{1}{2}(r)^2\left(\frac{\pi}{2}\right) \\ &= \frac{\pi}{4}r^2\end{aligned}$$

Calculating the area of the square considering the length of the square being r:

$$\begin{aligned}\mathbf{P} &= \frac{\pi}{4}r^2 / r^2 \\ \mathbf{P} &= \frac{\pi}{4}\end{aligned}$$

This proves that the selected quadrant, has a probabilistic area of $\frac{\pi}{4}$, when considering integrational approach, or straight geometrical approach.

5.1 Mathematical understanding.

The indicator variable I_k represents whether the k-th randomly chosen point lands in the quarter circle or not. Essentially it will be either 1 signifying that the point is inside, or it will be 0 showing that the point landed outside the quarter circle. As proven in the previous section, the theoretical probability of a point landing inside the unit square is $\frac{\pi}{4}$. The expectation of the indicator variable follows the probability.

$$I_k(x, y) = \begin{cases} 1 & , x^2 + y^2 \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

(Self-generated piecewise function using OverLeaf)

Since the probability of landing inside the quarter circle is same as the area:

$$P(I_k = 1) = p = \frac{\pi}{4}$$

Thus this gives me with the following formula:

$$E[I_k] = p = \frac{\pi}{4}$$

Regarding the estimation I must utilize a certain range of values for the constant N, as using N iterations. According to that, the sample mean can be determined as follows:

$$I_N = \frac{1}{N} \sum_{k=1}^N I_k$$

{By associating the variable I_k :}

$$\pi_N = 4I_N = \frac{4}{N} \sum_{k=1}^N I_k$$

Variance determination:

$$Var(I_k) = p(1 - p)$$

$$Var(I_N) = \frac{p(1-p)}{N}$$

$$Var(\pi_N) = \frac{16p(1-p)}{N}$$

After substituting $p = \frac{\pi}{4}$:

$$Var(\pi_N) = \frac{4\pi(4-\pi)}{N}$$

The standard deviation is :

$$\sigma_N = \sqrt{Var(\pi_N)}$$

$$\sigma_N = \sqrt{\frac{4\pi(4-\pi)}{N}}$$

$$\sigma_N = \sqrt{\frac{4\pi(4-\pi)}{10000}}$$

$$\sigma_N \approx 0.01642 \text{ (With 10,000 trials)}$$

The indicator variable I_k determines the outcome of the k^{th} sampled point, signifying whether it landed inside or outside the quarter. If it's a success it results as 1 meaning the point lands **inside** the quarter and if it's a failure it results as 0, and **not inside** the quarter.

The estimator $\pi_N = 4I_N$ logically follows from the sample mean probability, $p = \frac{\pi}{4}$, converting the geometric area relationship into a probabilistic approximation of π . The expression also contains π in the numerator, showing that the uncertainty in estimating π is inherently linked to the constant itself. 50 independent trials were conducted, each consisting of 10,000 random points. Hence this allows the behavior, distribution and stability of the Monte Carlo simulation to be examined across repeated experiments and trials.

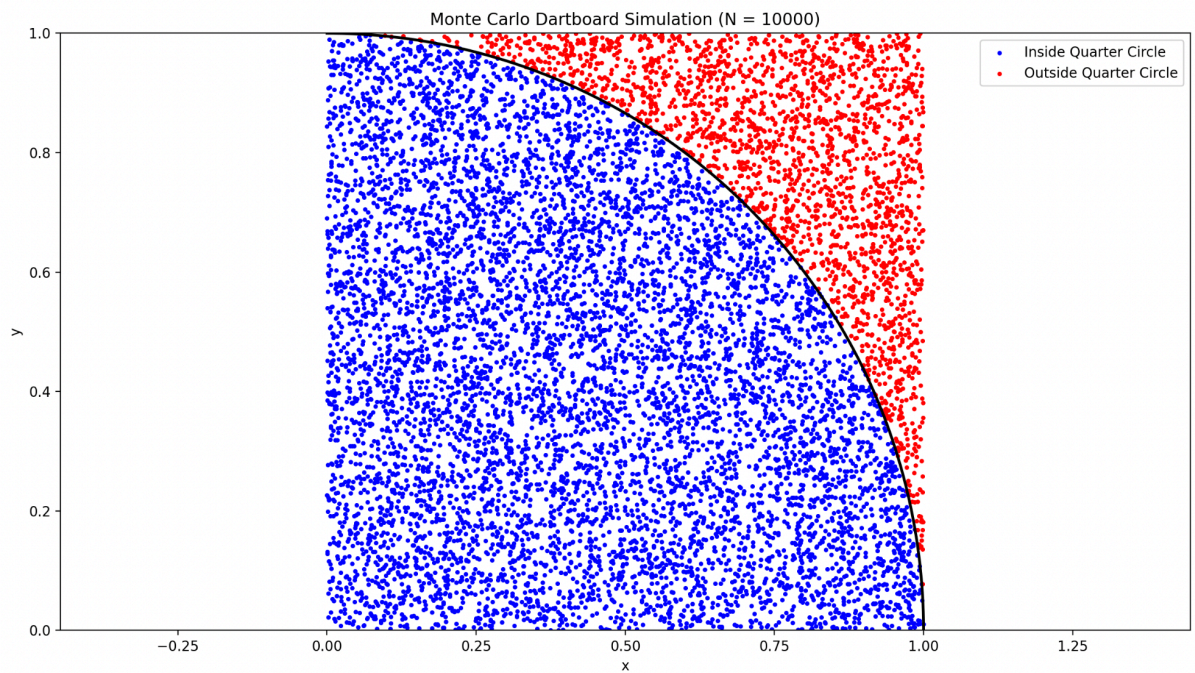
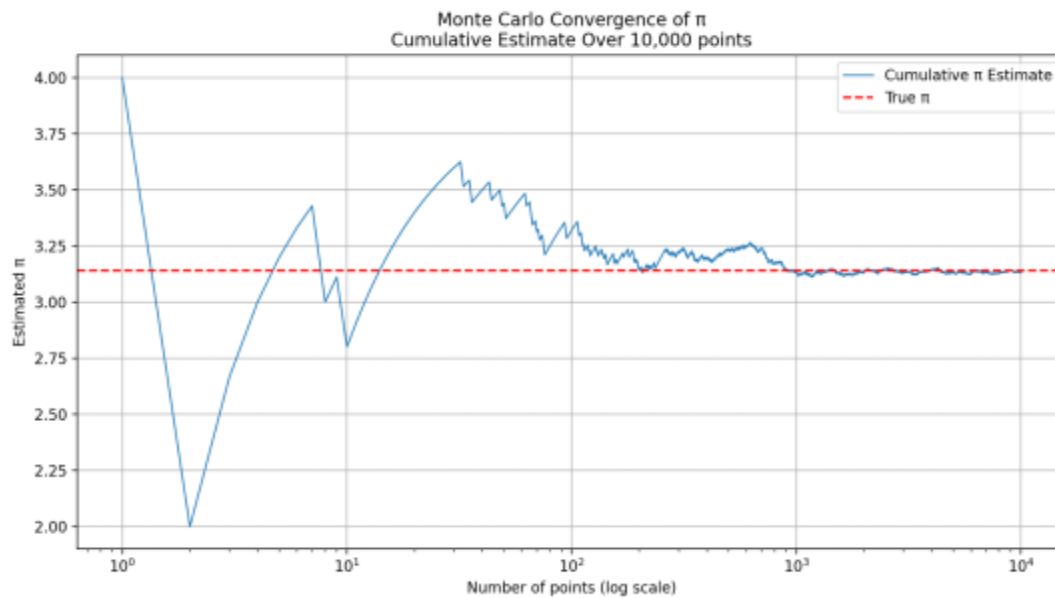
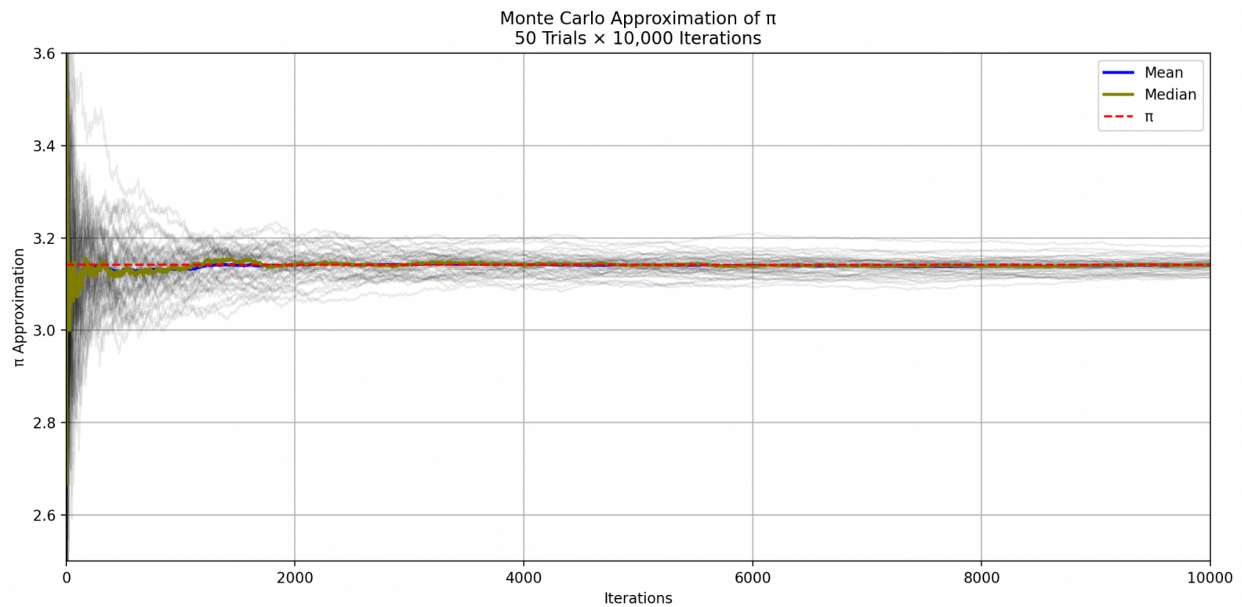


Figure 2 Monte Carlo Dartboard approximation with 10000 points (N) (Appendix 3)



Graph 3: Monte Carlo Single trial Convergence graph estimating towards π (Appendix 2)



Graph 4: Monte Carlo simulation graph (Appendix 2)

In the beginning we see that when the amount of points sampled are inaccurate and scattered. From the dartboard simulation shown in figure 2, we see that there are 10,000 points sampled. A larger concentration of points appears to be inside of the quarter circle, which reflects the probabilistic area of $\frac{\pi}{4}$. This repeats for every **50 trials** executed in order to determine an accurate result, each with 10,000 sampled points. This repetition is vital for analysing the statistical behavior of the estimator rather than relying on a single less probable and accurate random outcome.

All the 10,000 randomly general points, sampled randomly, some inside and outside the quarter. At the beginning of each trial, the estimator behaves very **chaotic** as only few points have been sampled. Across all the 50 trials, the first few hundred iterations show the largest variance.

As demonstrated in the graph above, the Monte Carlo approximation successfully converges towards π but slowly. During the early sample iterations, only a small number of points were successfully sampled, producing large variations in the estimation. As progressed ahead, the variance decreases mathematically, converging towards π . This behavior follows the variance expression derived earlier, $Var(\pi_N) = \frac{4\pi(4-\pi)}{N}$. Since the variance is inversely proportional to the number of samples, increasing N results in reduced overall variability of the estimator. As the variance decreases, the Monte Carlo estimate becomes progressively more stable and approaches π .

Therefore, by selecting N value as 10,000 iterations, it provides a sufficiently large sample size to reduce the variance and generate a statistically significant and meaningful approximation of π .

6.0 Comparative Analysis of Approximation Accuracy

6.1 Gregory - Leibniz Series Results in **tabular** format

N - Number of terms	Approximated π	Absolute Error
10	3.2323158094	0.0907231558
50	3.1611986130	0.0196059594
100	3.1514934011	0.0099007475
200	3.1465677472	0.0049750936
500	3.1435886596	0.0019960060
1000	3.1425916543	0.0009990007
2000	3.1420924037	0.0004997501
5000	3.1417926136	0.0001999600
10000	3.1416926436	0.0000999900

Table 1 (Appendix 1)

6.2 Monte Carlo Results in **tabular** format.

Trial	Approximated π	Absolute Error
5	3.1296	0.011993
10	3.1448	0.003207
15	3.1360	0.005593
20	3.1468	0.005207
25	3.1224	0.019193
30	3.1504	0.008807
35	3.1408	0.000793
40	3.1464	0.004807
45	3.1380	0.003593

Trial	Approximated π	Absolute Error
50	3.1604	0.018807

Table 2 (See Appendix 4 and 5)

The table 2 displays the final π estimated values produced by each of the 50 Monte Carlo trials. Each value slightly deviates from each other primarily due to randomness however the majority of the estimates lie between 3.12 and 3.17 demonstrating a chaotic or clustering around the true value of π . The absolute error differentiates across trials, reflecting the inherent variability of the Monte Carlo method. Essentially these results do not follow a monotonic pattern which is more frequent in the Gregory - Leibniz series.

6.3 Monte Carlo Summarized results

Statistical Type	Value
Mean estimate for π	3.139832
Median estimate for π	3.143400
Standard Deviation	0.016789
Minimum estimate	3.104000
Maximum estimate	3.170000

Table 3 (See Appendix 4)

In each of the tables 1 and 2, I have added an additional column that measures the absolute error separately for Gregory-Leibniz and Monte Carlo. To measure the accuracy the absolute error is generated for each value calculated. It quantifies how far an estimated value is from the true mathematical constant π , independent of whether the approximation lies above or below it. The absolute error is calculated with the following method:

$$|\pi_{Approx} - \pi|$$

Here the π_{Approx} represents the numerical approximation produced either by the Gregory - Leibniz partial sum or the Monte Carlo simulation. The constant π used is the actual determined irrational number. The reason for using the absolute value is to avoid positive and negative errors cancelling out. The Gregory-Leibniz partial sum alternate between overestimating and underestimating π due to the alternating nature of the series. Essentially, taking the magnitude of the difference ensures that the error reflects the size of the deviation, not its direction.

The results for the Gregory-Leibniz series approximation show a steady and a predictable decrease in the absolute error as the number of partial terms increases over time. This behavior perfectly aligns with the theoretical and the alternating structure of the series. As it is alternating, and is a conditionally convergent series, its error after certain sums, is confined by the magnitude of the next term. As we see in the following equation:

$$\left| \pi - S_N \right| \leq \frac{4}{2N+3}$$

This proves why the absolute error decreases consistently in the table and actually does not increase once N is large. Despite the fact that the convergence is really slow, the trend is essentially monotonic, meaning with more terms, the approximation improves. More importantly, the series is gradually approaching π asymptotically.

For the Monte Carlo approximation method, the results from the program provides a general decrease in the error calculation; however unlike Gregory - Leibniz, this trend isn't monotonic. As we know Monte Carlo simulation is a probabilistic approach rather than an alternating or monotonic approach, in the simulation, each approximation stage is heavily influenced by random sampling. In the beginning stages for small N values, the estimator gives a high variance result with inaccurate precision. Furthermore, the error reduction is noticeably slower than observed in the Gregory - Leibniz (table 1). This conversely corresponds to the theoretical variance equations derived above:

$$\begin{aligned} Var(\pi_N) &= \frac{4\pi(4-\pi)}{N} \\ \sigma_N &\propto \frac{1}{\sqrt{N}} \end{aligned}$$

Thereby since the standard deviation is inversely proportional to $\sqrt{N}$, it means that the Monte Carlo approach requires thousands of iterations before stabilising near π . When the error results are compared with the Gregory-Leibniz series' results, the Monte Carlo simulation converges far more slowly and unpredictably, highly due to the deterministic structure of the Gregory-Leibniz series.

Additionally the table 3 demonstrates the mean of the 50 trials lies close to the true and actual value of π , suggesting Monte Carlo simulations are theoretically unbiased, but finite simulations show small numerical bias.

7.0 Error analysis and theoretical bounds

The results in tables 2 and 3 demonstrate the effective convergence towards π , however we must evaluate the Monte Carlo method's limitations and potential refinements. The estimated π value as 3.139832 (table 3) suggests that there may be potential bias in the implementation. Essentially these limitations could be initiated due to the pseudo-random number generator or even floating-point precision errors throughout the computational representation.

The calculated standard deviation of 0.016789 quantifies the variability across the Monte Carlo trials, showing the probabilistic nature of the estimator. Because Monte Carlo simulations often rely on random sampling, the estimates exhibit inherent fluctuations, and do not converge

monotonically, like the Gregory-Leibniz series. Consequently, such methods are less suitable and reliable for ideas that require guaranteed precision. Variance reduction techniques such as stratified sampling, which divides the unit square into smaller regions to smaller segments, to ensure proportional sampling, which use paired samples to reduce variability, could thereby improve the stability of the estimator.

Additionally, **antithetic** variates as a method, could be used to produce point pairs to lower the overall variance while also maintaining the same amount of samples.

In contrast, the Gregory-Leibniz series follows a deterministic convergence pattern. The approximation error corresponds to the difference between the true value of π and the partial sum S_n . By the alternating series error bound, the magnitude of this error there satisfies:

$$|E_n| = |\pi - S_n| < |a_{n+1}|$$

When we are evaluating external mathematical theories such as the Gregory and Leibniz series, we must validate the results by using a generic approximation error method. The approximation error is primarily the difference between an exact value and its approximate or estimated value. The error is the difference between the true sum and the partial sum of the sequence in our context. To further be able to develop an accurate Gregory-Leibniz series in approximating the value of π , we must comply with the error term behavior as the number of n exponentially increases. This can successfully be determined by using the Alternating Series Test, and its corresponding error bound theorem, which provides a mathematical upper limit for the deviation of a finite partial sum from the actual value of the infinite series.

The error bound theorem mentions that for any convergent alternating series that satisfies the conditions in sections 3.2.1 and 3.2.2, the magnitude of the error after n terms is less than or equal to the magnitude of the next term.

$$|E_n| = |\Pi - S_n| < |a_{n+1}|$$

Since for the Gregory-Leibniz series:

$$a_{n+1} = \frac{4}{2(n+1)+1} = \frac{4}{2n+3}$$

The key element of an alternating series is that the error bound is very tight, so if the series meets the conditions for the alternating series test, the error is no larger than the absolute value of the first term that was not included in the partial sum. For this investigation I will be evaluating whether the true value of Π is correct by using the formula:

$$p_N = 4S_N = 4 \sum_{k=0}^N (-1)^k \frac{1}{2k+1}$$

The key strategy here will be that I will not be using the exact value of π , but instead try to model it directly by convergence and the error checking bounds. Therefore, the remainder after N terms will be determined by the following formula:

$$R_N = \frac{\Pi}{4} - S_N$$

8.0 Reflection and Conclusion

In order to produce an accurate approximation of π , it would be helpful to compare and contrast the Gregory-Leibniz alternating series and Monte Carlo simulation approaches with other established methodologies.

Gregory-Leibniz series is a converging series due to the negative and positive sums, and it is a slow converging series when it is compared to approaches like Machin-like formulas or the Machin's approach. Machin's, approach towards approximating π :

$$\frac{\pi}{4} = 4\arctan\left(\frac{1}{5}\right) - \arctan\left(\frac{1}{239}\right)$$

Machin's identity actually converges significantly faster because the Taylor expansion of $\arctan(x)$ converges rapidly when the x value is small. Since $\frac{1}{5}$ and $\frac{1}{239}$ are small values, the resulting series generates a much faster approximation, as compared to Gregory-Leibniz series. The use of $\arctan$ identity enables using the Taylor series and provides a more rapid convergence when compared to the Gregory - Leibniz method.

The Monte Carlo simulation investigated is fundamentally different, as it is a probabilistic estimator. The dartboard simulation approximates π by using random sampling, however its convergence is restricted by the statistical relationship $\sigma_N \propto \frac{1}{\sqrt{N}}$. Meaning that the accuracy would improve only with the square root of the sampled points, N .

Alternative probabilistic methods also exist, such as Buffon's Needle experiment, which estimates π using geometric probability and trigonometric relationships between the randomly dropped needles and parallel lines. Even though conceptually related with the dartboard simulation, Buffon's approach involves additional variables making the implementation mathematically complex.

In conclusion, based on the achieved results and graphical analysis indicate, that the Gregory-Leibniz series essentially provides a more stable, and predictable approximation of π compared to the Monte Carlo dartboard simulation. The series approach is deterministic and follows a structured alternating convergence pattern, which allows the approximation to finally improve systematically as the number of terms increases. In contrast, the Monte Carlo approach is probabilistic, and heavily relies on random sampling of points within the unit square, which introduces variability between trials. This randomness further produces fluctuations in the estimator, and increases the variance when the number of samples is small. Although Monte Carlo's accuracy improves as the iterations increase, it requires a significantly larger amount of samples to reach the true value of π . Therefore while both methods approximate π , the Gregory-Leibniz series shows a more consistent and reliable convergence behavior.

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Appendices

Appendix 1 :- Code for the Alternating Gregory-Leibniz series graph

```
import os, math, csv
import numpy as np
import matplotlib.pyplot as plt

N_MAX_PLOT    = 2000
N_MACHIN      = 120
MAX_ITER_EMP  = 100000

def empiricaltermsfor_decimals(d, max_iter=100000):
```

```

    thresh = 0.5 * (10 ** (-d))
    s = 0.0
    for n in range(max_iter + 1):
        s += ((-1.0) ** n) / (2*n + 1)
        if abs(math.pi - 4.0*s) < thresh:
            return n
    return None

# Gregory-Leibniz partial sums
ns = np.arange(0, N_MAX_PLOT + 1)
terms = ((-1.0) ** ns) / (2 * ns + 1)
S_partial = 4.0 * np.cumsum(terms)
pi = math.pi
errors = np.abs(S_partial - pi)

csv_path = os.path.join(OUT_DIR, "leibnizpartialsumsTable.csv")
with open(csv_path, "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(["N", "S_N", "abs_error"])
    for n, s, e in zip(ns, S_partial, errors):
        w.writerow([int(n), float(s), float(e)])

# Gregory - Leibniz Convergence graph
plt.figure(figsize=(8, 4.5))
plt.step(ns, S_partial, where="post", linewidth=1.5)
plt.axhline(pi, linestyle="--", linewidth=1.5, label=r"$\pi$")
plt.legend(); plt.tight_layout()
plt.savefig(os.path.join(OUT_DIR, "leibniz_zigzag.png"), dpi=200)

even_idx = ns[ns % 2 == 0]
odd_idx = ns[ns % 2 == 1]
S_even = S_partial[even_idx]
S_odd = S_partial[odd_idx]

plt.figure(figsize=(8, 4.5))
plt.plot(even_idx, S_even, label="Even partial sums")
plt.plot(odd_idx, S_odd, label="Odd partial sums")
plt.axhline(pi, linestyle="--", linewidth=1.5)
plt.legend(); plt.tight_layout()

```

```

plt.savefig(os.path.join(OUT_DIR, "leibniz_even_odd.png"), dpi=200)

bound = 4.0 / (2 * ns + 3)
mask = ns >= 1

plt.figure(figsize=(8, 4.5))
plt.plot(ns[mask], errors[mask], linewidth=1.5)
plt.plot(ns[mask], bound[mask], linestyle="--", linewidth=1.5)
plt.yscale("log")
plt.legend(); plt.tight_layout()
plt.savefig(os.path.join(OUT_DIR, "leibniz_error_bound.png"), dpi=200)

N_values = [10, 50, 100, 200, 500, 1000, 2000, 5000, 8000, 10000]

print("\n6.1 Gregory-Leibniz Results (Tabular Format):")
print("Absolute Error")

// determined row array for GL series:
GL_rows = []

for N in N_values:
    ns_local = np.arange(0, N+1)
    terms_local = ((-1)**ns_local) / (2*ns_local + 1)
    approx_pi = 4*np.sum(terms_local)
    err = abs(math.pi - approx_pi)
    GL_rows.append([N, approx_pi, err])
    print(f"{N}\t{approx_pi:.10f}\t{err:.10f}")

with open("GL_results_table.csv", "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(["N", "Approx_pi", "Absolute_error"])
    for row in GL_rows:
        w.writerow(row)

```

Appendix 2 :- Code for Monte Carlo Convergence graph towards π .

```

import numpy as np
import matplotlib.pyplot as plt

N = 10000

x = np.random.rand(N)
y = np.random.rand(N)
inside = (x*x + y*y <= 1)

cumulative_hits = np.cumsum(inside)
trials = np.arange(1, N+1)
pi_estimates = 4 * cumulative_hits / trials

plt.figure(figsize=(12,6))
plt.plot(trials, pi_estimates, label="Cumulative  $\pi$  Estimate", linewidth=1)
plt.axhline(np.pi, color='red', linestyle='--', label="True  $\pi$ ")

plt.xscale("log")
plt.xlabel("Number of points (log scale)")
plt.ylabel("Estimated  $\pi$ ")
plt.title("Monte Carlo Convergence of  $\pi$ \nCumulative Estimate Over 10,000 points")
plt.grid(True)
plt.legend()
plt.show()

```

Appendix 3 :- Monte Carlo Dartboard simulation

```

import numpy as np
import matplotlib.pyplot as plt

# follows same N values as the convergenceGraph.py)
N = 10000
x = np.random.rand(N)
y = np.random.rand(N)
inside = x*x + y*y <= 1
# DARTBOARD PLOT
plt.figure(figsize=(7, 7))

plt.scatter(x[inside], y[inside], s=5, color='blue', label="Inside Quarter Circle")
plt.scatter(x[~inside], y[~inside], s=5, color='red', label="Outside Quarter Circle")

```

```

# Draw the quarter arc boundary
theta = np.linspace(0, np.pi/2, 300)
plt.plot(np.cos(theta), np.sin(theta), color='black', linewidth=2)

plt.title(f"Monte Carlo Dartboard Simulation (N = {N})")
plt.xlabel("x")
plt.ylabel("y")
plt.axis("equal")
plt.legend()
plt.xlim(0, 1)
plt.ylim(0, 1)

plt.show()

```

Appendix 4 :- Code for providing 50 trials and results for Monte Carlo Approach.

```

import numpy as np
import pandas as pd
import math

T = 50 #50 trials
N = 10000 #Each of the 50 trials, has 10000 points
results = []

for trial in range(1, T + 1):
    x = np.random.rand(N)
    y = np.random.rand(N)
    inside = (x*x + y*y <= 1)

    pi_est = 4 * inside.mean()
    abs_error = abs(pi_est - math.pi)

    results.append([trial, pi_est, abs_error])

```



```

df = pd.DataFrame(results, columns=["Trial", "Pi_Estimate", "Absolute_Error"])

df["Cumulative_Mean"] = df["Pi_Estimate"].expanding().mean()
df["Cumulative_SD"] = df["Pi_Estimate"].expanding().std()

df.to_csv("MC_50_trials_results.csv", index=False)

# prints all the 50 rows
print(" FULL MONTE CARLO TABLE (50 TRIALS) ")
print(df.to_string(index=False))

print(" summary ")
print(f"Mean  $\pi$  estimate      = {df['Pi_Estimate'].mean():.6f}")
print(f"Median  $\pi$  estimate      = {df['Pi_Estimate'].median():.6f}")
print(f"Standard deviation      = {df['Pi_Estimate'].std():.6f}")
print(f"Minimum estimate        = {df['Pi_Estimate'].min():.6f}")
print(f"Maximum estimate         = {df['Pi_Estimate'].max():.6f}")

```

Appendix 5 :- Complete Tabular Data for 1 to 50 trials connected with Appendix 4.

Trial	Pi_Estimate	Absolute_Error	Cumulative_Mean	Cumulative_SD
1	3.144	0.002407	3.144	Not defined
2	3.1172	0.024393	3.1306	0.01895
3	3.1304	0.011193	3.130533	0.0134
4	3.1176	0.023993	3.1273	0.01271
5	3.1296	0.011993	3.12776	0.011055
6	3.1264	0.015193	3.127533	0.009903
7	3.128	0.013593	3.1276	0.009042
8	3.1308	0.010793	3.128	0.008447
9	3.1144	0.027193	3.126489	0.00911
10	3.1448	0.003207	3.12832	0.010359

11	3.1524	0.010807	3.130509	0.012218
12	3.1132	0.028393	3.129067	0.012676
13	3.1668	0.025207	3.131969	0.016025
14	3.1492	0.007607	3.1332	0.016071
15	3.136	0.005593	3.133387	0.015503
16	3.1448	0.003207	3.1341	0.015247
17	3.128	0.013593	3.133741	0.014836
18	3.15	0.008407	3.134644	0.014895
19	3.1544	0.012807	3.135684	0.015168
20	3.1468	0.005207	3.13624	0.014971
21	3.1068	0.034793	3.134838	0.015944
22	3.1188	0.022793	3.134109	0.015931
23	3.1636	0.022007	3.135391	0.016735
24	3.1496	0.008007	3.135983	0.016622
25	3.1224	0.019193	3.13544	0.016498
26	3.158	0.016407	3.136308	0.016759
27	3.1588	0.017207	3.137141	0.016994
28	3.118	0.023593	3.136457	0.017064
29	3.146	0.004407	3.136786	0.01685
30	3.1504	0.008807	3.13724	0.016743
31	3.1672	0.025607	3.138206	0.017318
32	3.1384	0.003193	3.138212	0.017037
33	3.146	0.004407	3.138448	0.016823
34	3.1488	0.007207	3.138753	0.016661
35	3.1408	0.000793	3.138811	0.016418
36	3.1312	0.010393	3.1386	0.016231
37	3.1368	0.004793	3.138551	0.016007
38	3.146	0.004407	3.138747	0.015835
39	3.17	0.028407	3.139549	0.016407
40	3.1464	0.004807	3.13972	0.016232
41	3.104	0.037593	3.138849	0.016971
42	3.1448	0.003207	3.13899	0.016788
43	3.12	0.021593	3.138549	0.016838
44	3.1688	0.027207	3.139236	0.017254
45	3.138	0.003593	3.139209	0.017058

46	3.13	0.011593	3.139009	0.016922
47	3.1428	0.001207	3.139089	0.016746
48	3.1336	0.007993	3.138975	0.016586
49	3.1604	0.018807	3.139412	0.016695
50	3.1604	0.018807	3.139832	0.016789